

DAY-2:

Tasks

1.

List 3 real-world devices in your home or college that use SMPS (Switched Mode Power Supply) and 2 devices that use UPS (Uninterruptible Power Supply), explaining in one line why each device needs that type of power supply.

Device	Type of Power Supply	Why it needs it
Desktop PC	SMPS	Converts AC power from the wall socket into stable DC voltages required by the computer components.
LED Monitor	SMPS	Converts AC electricity into the low-voltage DC power required by the monitor's electronic circuits.
Laptop Charger	SMPS	Converts AC power into regulated DC power to charge the laptop battery and power the laptop.

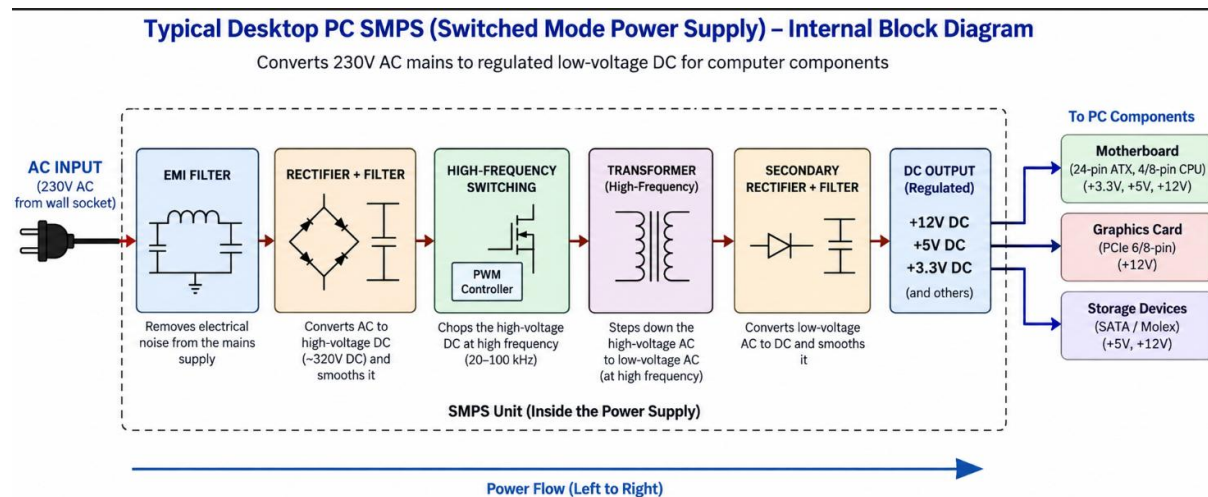
Device	Type of Power Supply	Why it needs it
Desktop Computer	UPS	Provides backup power during power cuts and protects the computer from sudden power fluctuations.
Wi-Fi Router	UPS	Keeps the router working during short power outages so the internet connection remains available.

2.

Draw and label a diagram showing the main internal components of a typical SMPS unit found in a desktop PC, including AC input, rectifier, transformer, and DC output

sections.

Hint: You can hand-draw and upload a photo, or use any free diagram tool like draw.io.

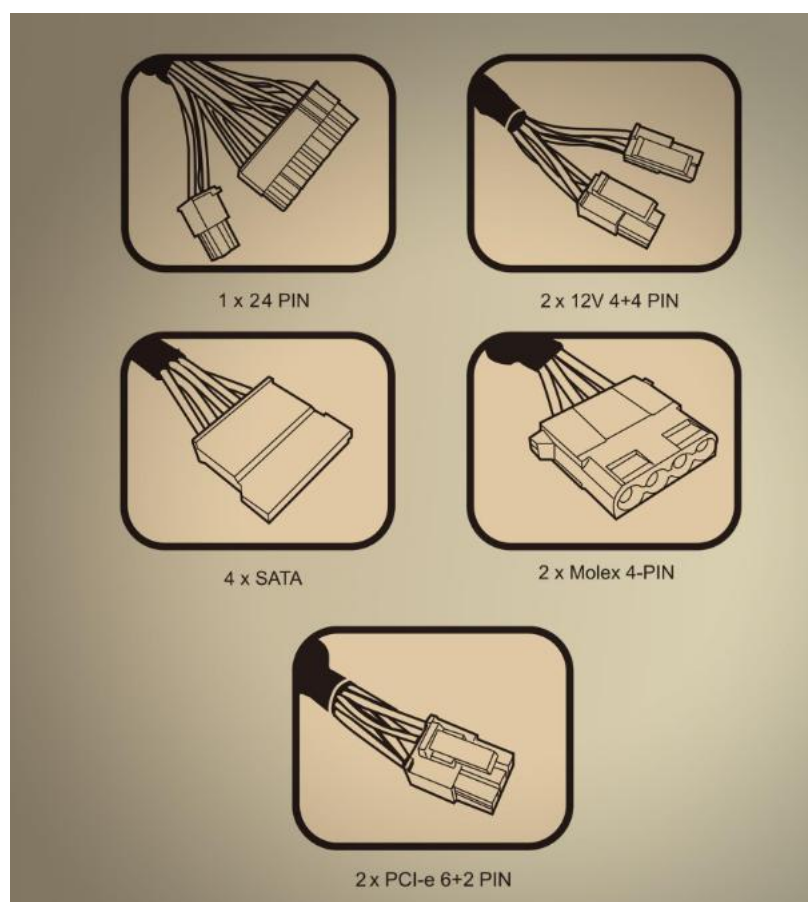


3.

Take photos or find clear images online of the following power connectors: ATX 24-pin motherboard connector, SATA power connector, Molex connector, and PCIe 6/8-pin connector. For each, write its name, the devices it powers, and one safety tip for connecting/disconnecting it.



1. 24-Pin Motherboard Connector
2. 4-Pin Molex Connector
3. PCI-E 6-Pin Connector
4. PCI-E 8-Pin Connector
5. SATA Power Connector
6. +12V 8-Pin Motherboard Connector
7. +12V 4-Pin Motherboard Connector



1. ATX 24-Pin Motherboard Connector

- Name: ATX 24-pin connector
- Powers: Motherboard and its major circuits.
- Safety tip: Turn off and unplug the PSU before connecting or disconnecting it. Make sure the locking clip is properly aligned and the connector is fully seated.

2. SATA Power Connector

- Name: SATA power connector
- Powers: SATA SSDs, HDDs, and some fan/RGB controllers.
- Safety tip: Do not force the connector; it is keyed and should slide into the socket in only the correct orientation.

3. Molex Connector

- Name: 4-pin Molex connector
- Powers: Older hard drives, optical drives, case fans, and some older peripherals.
- Safety tip: Switch off and unplug the PC before connecting or removing it, and pull the connector straight rather than yanking the wires.

4. PCIe 6/8-Pin Connector

- Name: PCIe 6-pin / 6+2-pin connector
- Powers: Graphics cards and other PCIe devices requiring additional power.
- Safety tip: They are not interchangeable.

4.

Imagine you are setting up a new gaming PC. List the steps you would follow to connect the SMPS to the motherboard, graphics card, and storage devices, mentioning which specific cables and connectors are used for each.

Step 1: Install the SMPS

- Place the SMPS inside the PC case.
- Secure it with the screws provided.
- Do not connect the AC power cable yet.

Step 2: Connect the motherboard

Use the 24-pin ATX cable from the SMPS.

SMPS - 24-pin ATX cable -> Motherboard 24-pin socket

This supplies the main power to the motherboard.

Step 3: Connect CPU power

Use the 4+4-pin EPS/ATX12V cable.

SMPS- 4+4-pin EPS cable -> CPU power socket on motherboard

Depending on the motherboard, this may be an 8-pin or 8+4-pin CPU power connection. Do not substitute a PCIe cable for the CPU/EPS cable.

Step 4: Connect the graphics card

If the graphics card requires additional power, use the appropriate PCIe 6-pin or 6+2-pin cable.

SMPS -PCIe 6+2-pin cable -> Graphics Card

Some newer graphics cards instead use a 12V-2x6 (12+4-pin) connector, so follow the GPU and PSU manufacturer's instructions.

Step 5: Connect storage devices

For SATA SSDs or HDDs, use the SATA power cable.

SMPS - SATA power cable -> SSD / HDD

The SATA data cable is a separate cable that connects the drive to the motherboard.

Step 6: Connect other peripherals

If an older device requires a 4-pin Molex connector, connect it using the PSU's peripheral/Molex cable.

Step 7: Check all connections

- Make sure every connector is fully seated.
- Check that cables are not trapped in fans.
- Make sure the correct cable is connected to each component.
- For a modular PSU, verify each cable is connected to the correct labelled PSU port.

Step 8: Connect AC power

Finally, connect the PSU's AC power cable to the wall socket or UPS and switch the PSU on.